

Computational Frameworks for Space Combustion: PDE-based Flame Synthesis and Optimal Control in Microgravity Environments

Cartik Sharma

Neuromorph Labs, Space Physics Division

Abstract

Modeling combustion in space requires solving complex thermal-chemical partial differential equations (PDEs) coupled with optimal control theory for propellant management. Here we present an integrated simulation environment that resolves the 1-D premixed laminar flame front using Arrhenius kinetics and finite-difference methods. Furthermore, we implement Pontryagin's Minimum Principle for fuel-optimal throttle velocity and analyze vehicle dynamics using linearized finite math in state-space form. Our results provide high-fidelity signatures of combustion efficiency and trajectory stability for next-generation orbital platforms.

1. Introduction

Space propulsion relies on the precise management of chemical energy release. In microgravity, the absence of buoyancy-driven convection alters flame topology, making diffusive transport dominant. To predict these signatures, robust computational frameworks are required that can handle stiff PDEs and provide real-time optimal control for rocket trajectories.

2. Methods: Partial Differential Equations

The simulation resolves the evolution of temperature (T) and species mass fractions (Y_F , Y_O) governed by the following system of non-linear PDEs:

$$dT/dt = \alpha * d^2T/dx^2 + (Q * w) / (\rho * C_p)$$

$$dY_F/dt = D * d^2Y_F/dx^2 - w / \rho$$

$$dY_O/dt = D * d^2Y_O/dx^2 - (nu * w) / \rho$$

Where w represents the Arrhenius reaction rate: $w = A * (\rho * Y_F) * (\rho * Y_O) * \exp(-E_a / RT)$.

3. Optimal Throttle and Finite Math

To minimize fuel consumption during ascent, we employ Pontryagin's Minimum Principle. The dynamics are modeled in state-space form:

$$\mathbf{dx}/dt = \mathbf{Ax} + \mathbf{Bu}$$

Where $\mathbf{x}$ is the state vector [altitude, velocity, pitch, pitch_rate]^T. Stability is determined by the eigenvalues (λ) of the Jacobian matrix $\mathbf{A}$. The state transition matrix is computed as:

$$\Phi(\mathbf{t}) = \exp(\mathbf{At}) = \mathbf{I} + \mathbf{At} + (\mathbf{At})^2/2! + \dots$$

4. Results and Continued Fractions

Our PDE solver successfully captures the flame front for H₂-O₂ and RP1-LOX fuels. For H₂-O₂ at 1 atm, we observe a peak temperature of 2800K and a laminar flame speed of 2.5 m/s. Additionally, we use continued fractions to compute orbital resonance ratios. For the constant π , the convergent 355/113 provides a precision of 2.6e-7, essential for high-frequency orbital calculations.

5. Conclusion

The integrated Space Combustion Physics platform demonstrates the synergy between PDE simulation, finite math dynamics, and optimal control. This framework serves as a prerequisite for more complex 3-D simulations of combustion in future deep-space exploration missions.